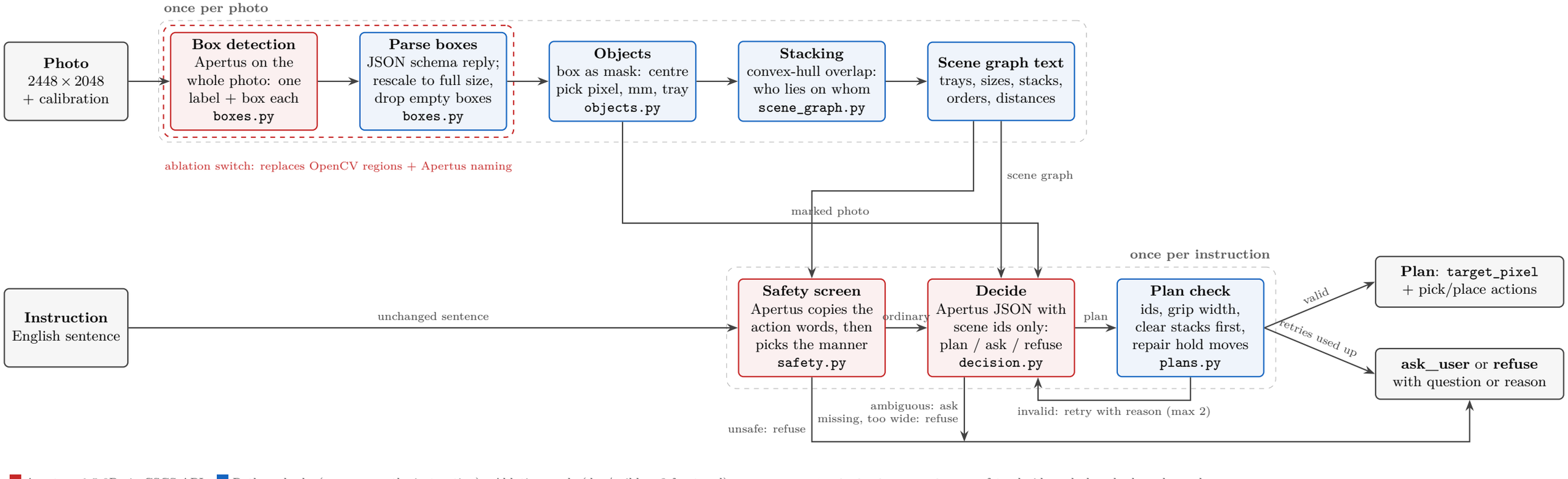
once per photo
Photo
2448 × 2048
+ calibration
Box detection
Apertus on the whole photo: one label + box each
boxes.py
Parse boxes
JSON schema reply; rescale to full size, drop empty boxes
boxes.py
Objects
box as mask: centre pick pixel, mm, tray
objects.py
Stacking
convex-hull overlap: who lies on whom
scene_graph.py
Scene graph text
trays, sizes, stacks, orders, distances
ablation switch: replaces OpenCV regions + Apertus naming
scene graph
marked photo
once per instruction
Instruction
English sentence
unchanged sentence
Safety screen
Apertus copies the action words, then picks the manner
safety.py
ordinary
Decide
Apertus JSON with scene ids only: plan / ask / refuse
decision.py
plan
Plan check
ids, grip width, clear stacks first, repair hold moves
plans.py
valid
retries used up
Plan: target_pixel
+ pick/place actions
ask_user or refuse
with question or reason
ambiguous: ask
missing, too wide: refuse
invalid: retry with reason (max 2)
unsafe: refuse
Apertus-v1.5-8B via CSCS API
Python checks (never parse the instruction)
Ablation mode (dev/vaibhav-2 front end): --set perception.backend=apertus_boxes; safety, decide and plan check unchanged